

Silo-bag system for storage of grains, seeds and by-products: A review and research agenda

Ricardo Bartosik^{a,*}, Hernán Urcola^a, Leandro Cardoso^a, Gisele Maciel^a, Patrizia Busato^b

^a Instituto de Innovación para la Producción Agropecuaria y el Desarrollo Sostenible (IPADS) Balcarce (INTA-CONICET), RN 226 km 73.5, Balcarce, Buenos Aires, Argentina

^b Department of Agriculture, Forestry and Food Sciences (DISAFA), Faculty of Agriculture, University of Turin, Largo Braccini 2, 10095, Grugliasco, Italy

ARTICLE INFO

Keywords:

Hermetic storage
Grain
Oilseeds
Logistics

ABSTRACT

Silo-bags (grain bag, sausage bag or silo bolsa) have been used for storing grains in Argentina for more than 25 years, and are now fully integrated in the grain postharvest system at different levels with important economic and logistic benefits for the agricultural sector. Additionally, silo-bags are being adopted in more than fifty countries, from the tropics to the cold regions of the world. Given the interest that silo-bag technology is arousing worldwide, it is timely to review the state of the art and identify gaps in scientific knowledge. The scope of this review includes the particular ecosystem of the grain stored in silo-bags, the abiotic components (temperature, moisture content and gasses) and interactions with biotic components (microorganisms and insects), alternatives for insect control treatments, effect of storage conditions on grain quality, modelling work, and the impact of the silo-bag system on the economics and logistics of the agricultural sector. Additionally, gaps in knowledge that need to be addressed in future research projects are identified and discussed.

1. Introduction

Silo-bag (grain bag, sausage bag or silo bolsa) is a grain storage system made with a plastic bag, shaped like a tube of variable length and diameter (most typically 60 m long and 2.7 m diameter), with a holding capacity of about 200 metric tonnes (T) of corn, wheat or soybean. The plastic cover is usually white on the outside, black on the inside and it is used for storing different grains and by-products during a variable period of time, from a few weeks to several months (Bartosik, 2012). In Argentina, the use of silo-bags for storing dry grains evolved from their use for anaerobic storage of chopped forages in the mid '90s (Casini, 2014). The first bags were imported and the bagging and unloading operations were carried out with adapted technologies. The silo-bag system, as it is known today, is made up of different specifically developed technological components including the bagging machine, the grain cart with unloading auger (or truck), the plastic bag and the unloading equipment (Santa Juliana and Cardoso, 2014). These components were developed in the late '90s and the beginning of the 2000s by private companies in collaboration with the National Institute of Agricultural Technology (INTA) of Argentina. These innovations, together with a strong extension program financed by the public and private sectors and implemented by INTA, led to the massive adoption of

the silo-bags by the entire agricultural sector of the country, including farmers, grain elevators, grain processing industries and ports. In 1997 less than 1 million (T) per year of different grains were stored in silo-bags in Argentina, a few years later in 2003 this figure increased to more than 10 million T, and since 2015 between 40 and 55 million T are being stored (based on compiled information from the Argentine silo-bags manufacturing companies). Today, silo-bags are used for storing feed grains, industrial grains, specialty grain, organic grains, and various by products (Alvarez et al., 2016; Bartosik et al., 2008b, 2012; Cardoso et al., 2014; Chelladurai et al., 2016; Faroni et al., 2009). In addition to their extensive adoption in Argentina, silo-bags are used in more than 50 countries under very different climatic conditions, from the tropics (e.g. Brazil, Colombia, Ecuador, Central America, sub-Saharan countries, among others) to regions with a temperate climate (e.g. Australia, South Africa, Chile, United States, Turkey, among others), and even regions with cold climates (e.g. Canada, Russia, Ukraine, among others) (CIDETER, 2022).

In almost 30 years of use of silo-bags for grain storage an important volume of scientific information has been generated. However, this information has not been properly systematized and analyzed. Therefore, the objectives of this work were to critically review the state of the art related to silo bag technology and to identify gaps in the current

* Corresponding author.

E-mail address: bartosik.ricardo@inta.gob.ar (R. Bartosik).

<https://doi.org/10.1016/j.jspr.2022.102061>

Received 27 September 2022; Received in revised form 1 December 2022; Accepted 6 December 2022

Available online 14 December 2022

0022-474X/© 2022 Elsevier Ltd. All rights reserved.

knowledge for future research initiatives.

2. Silo-bag ecosystem

The typical biotic components of the stored grain ecosystem are insects, microorganisms and the grain or seed. In the presence of oxygen (O_2), the main metabolic pathway used by biotic agents to obtain energy is aerobic respiration, through which glucose is oxidized to carbon dioxide (CO_2) (releasing also moisture and heat). The respiration rate of the grain bulk is affected by grain type (Jian et al., 2014; Ochandio et al., 2014a), grain condition (presence of foreign matter, mechanical damage, and microbial load) (Al-Yahaya et al., 1993; Bern et al., 2002) and its interaction with the abiotic component of the system: moisture content (m.c.) and equilibrium relative humidity (r.h.), temperature, and internal gas composition (Marcos Valle et al., 2021b; Ochandio et al., 2017). Additionally, the presence of insects also contributes to the overall respiration of the bulk (Chotikasatien et al., 2017; Cofie-Agblor et al., 1995; Sone, 1999). During storage, temperature, m. c., microbial load, and O_2 concentration vary affecting respiration, whose cumulative effect is usually quantified as dry matter loss (d.m.l.) (Steele et al., 1969; Thompson, 1972).

a. Grain temperature

The temperature of the grain stored in the bag at a given time depends on the temperature of the grain at the time of bagging (this effect is less important the longer the time that elapses after bagging), solar radiation, heat release from the respiration of biotic components, and heat transfer with the air surrounding the bag and the soil where the bag is laying (Gastón et al., 2009). Several authors showed that the temperature at the top grain layer (0.1 m depth) follows the distinctive pattern of the ambient air temperature, reaching its maximum at noon and minimum during the early morning (Bartosik et al., 2008b; Ward and Davis, 2013, among others). The daily temperature oscillation decreased with the grain depth, being unnoticeable at 0.4 m from the surface (Darby and Caddick, 2007). The relationship between the grain and ambient temperatures was characterized in several field experiments with different grains and under different climates (Bartosik et al., 2008b; Chelladurai et al., 2016b; Darby and Caddick, 2007; Ochandio et al., 2010; Ward and Davis, 2013), showing that the bulk grain temperature decreases during storage from summer to winter and increases from winter to summer, trending to equilibrate with the ambient temperature of the season (Fig. 1). The grain temperature at the top layer could result higher than the ambient temperature due to solar radiation, with potential negative effects on grain quality (spoiled grain and heat damage).

In comparison with the storage in standard bins (metal or concrete), where the grain temperature is mainly controlled by means of forced

aeration, the possibilities of external control of the grain temperature in silo-bags are very limited. Clemente et al. (2009) covered the silo-bag with shade cloth to reduce solar radiation, achieving a reduction of 4 °C in the grain temperature. However, this is not a current practice and the bags are usually placed in the field without any cover. Despite this, the temperature of the grain during storage in the silo-bag does not rise to unsafe levels (to our knowledge there is no report about grain combustion process in silo-bags), even with slightly moist grain, as it would if the grain were stored in a metal bin. A reason for this is that a standard silo-bag (200 t capacity, 60 m length and 2.7 m diameter) has a surface/volume ratio of 1.4 m²/m³, almost double than that of a regular bin of similar capacity (0.78 m²/m³ for a 7 m diameter and 9 m height bin of 200 T of capacity). Also the shorter distance between the center of the grain mass and the surface (0.8 m for the central-longitudinal axis of the bag to the soil or the air for silo-bag and more than 3 m for a standard storage bin), facilitates a natural heat exchange of the grain mass with the surrounding ambient (air and soil), dissipating the heat of respiration. Additionally, if the storage conditions are such that the O_2 concentration is substantially decreased, the respiration rate (and heat released) is also diminished (Marcos Valle et al., 2021b; Ochandio et al., 2017). The underlying mechanisms involved in the evolution of stored grain temperatures described above were confirmed through modelling by several authors (Arias Barreto et al., 2013a, 2017; Gastón et al., 2008, 2009; Jian et al., 2015a).

b. Moisture

There is no water vapor permeability coefficients reported for silo-bags films. As reference, a permeance of 16.17 g m⁻² d⁻¹ atm⁻¹ was derived for a low density polyethylene (LDPE) film 230 µm thick (Gaikwad and Rubino, 2017). From a practical point of view, this water vapor permeance level is sufficiently low to prevent a measurable grain moisture variation during storage. Thus, unless rainwater enters to the bag through openings, the average bulk grain m. c. Should remain constant.

Ward and Davis (2013) measured variations in temperature and equilibrium r. h. at different locations of the silo-bag and found a similar pattern in the daily oscillation of these two variables. Near the surface of the bag the daily equilibrium r. h. oscillation was very noticeable, but towards the center of the bag this variable remained constant. Jian et al. (2015a,b) and Gastón et al. (2009) modeled the heat and moisture transfer during grain storage in silo-bags and concluded that the temperature differential between the top layer and the rest of the bag caused migration of moisture from the core of the grain mass to the top layer, and, to a lesser extent, to the bottom layer, which has been confirmed by several field trials. The magnitude of the moisture stratification process during silo-bag storage is currently not clear. Gastón et al. (2009) considered that grain m. c., temperature fluctuation magnitude and

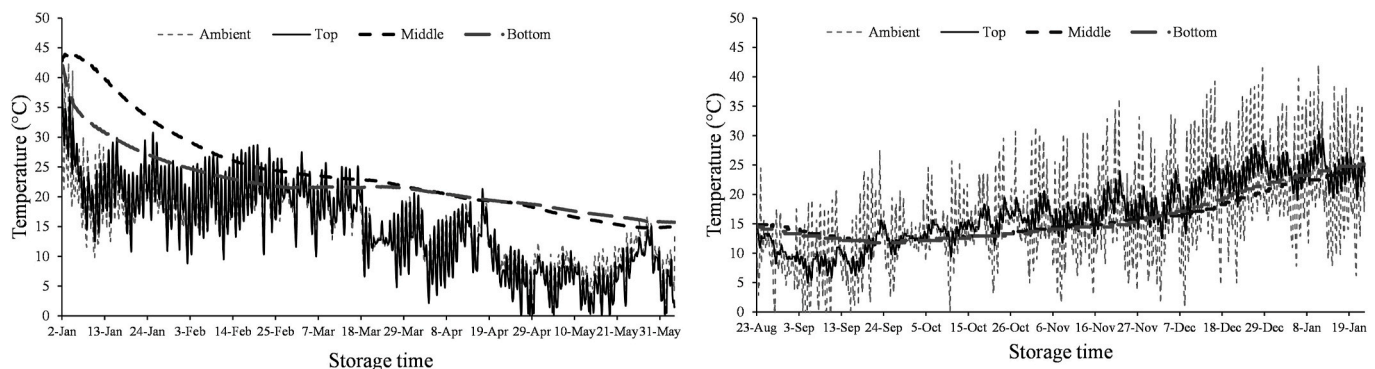


Fig. 1. Evolution of ambient and grain temperatures (wheat in the left and corn in the right) at different layers of a silo-bag (top: 0.05 m from the surface, middle: 0.70 m and bottom: 1.50 m) from January (summer) to May (winter) in the left, and from August (winter) to January (summer) in the right. Adapted from (Bartosik et al., 2008b).

storage time determine the magnitude of moisture stratification. This is of importance because severe moisture migration may increase the risk of grain spoilage (and grain quality deterioration) in localized areas. Darby and Caddick, (2007) reported some moisture stratification during storage of dry barley ($\leq 11\%$ m. c.) under Australian conditions. This stratification increased m. c. in the peripheral grain layer up to 13% over winter, but remained dry over summer with temperatures above 30 °C, indicating that the grain could be stored in good condition for up to 6 months. However, Butts and Ward (2022) reported a severe moisture migration in the upper layer (15 cm deep) of a peanut silo-bag. Coincidentally, Bartosik et al. (2008) also reported a substantial moisture increase in the top layer of a sunflower silo-bag (16% average m. c.). This would indicate that slightly wet (high equilibrium r. h. conditions) and hygroscopic grains would be more prone to develop the moisture migration phenomena. On the other hand, empirical evidence (Bartosik et al., 2008b; Cardoso et al., 2010; Ochandio et al., 2010) and simulation studies (Gastón et al., 2009) indicated that the storage of dry grain (with equilibrium r. h. lower than 65%) does not lead to perceptible moisture migration in a storage period of up to one year, even in hygroscopic materials such as sunflower pellets (Behr et al., 2021).

c. Internal atmosphere

Under absolute airtight conditions, the respiration of biotic components of bulk grain tends to consume all the O_2 , generating an anoxic atmosphere enriched with CO_2 (Weinberg et al., 2008). However, silo-bags are not completely airtight (Cardoso et al., 2012), thus the internal gases concentration depends on the balance between the respiration rate (O_2 consumption and CO_2 generation) and the gas exchange rate with ambient air (entrance of O_2 to the silo-bag and the exit of CO_2 to the ambient air) (Fig. 2). The movement of gases in and out of silo-bags depends on the gas partial pressure differential and the overall effective permeability of the silo-bag, taking into account the permeance of the plastic film and the contribution of perforations (Bartosik, 2012). Thus, the internal gases concentration, at any given storage time, is the result of the equilibrium between the respiration and gasses exchange rates. This equilibrium is dynamic due to variations in both, respiration

and gas exchange rates. Respiration rate depends mainly on the moisture availability, characterized as water activity (a_w) (related to the concept of equilibrium r. h.), and the temperature of the grain bulk (Ochandio et al., 2017). Thus, respiration can vary during storage due to the fluctuating bulk temperature through the seasons, rainwater ingress through perforations in the plastic cover, or condensation buildup in the peripheral grain layer, among others. The gas exchange rate, on the other hand, can vary (in general increase) due to the accumulation of perforations in the plastic cover of the silo-bag (Cardoso et al., 2012). As a consequence, the concentrations of O_2 and CO_2 inside a silo-bag change throughout storage time (Arias Barreto et al., 2013b; Cardoso et al., 2010; Ochandio et al., 2010). However, grain m. c. seems to be the most influential factor in gas concentration (Cardoso et al., 2008; Rodríguez et al., 2008). Fig. 3 plots the observations of Rodríguez et al. (2008) showing that for dry grain conditions (m.c. below 14%) O_2 concentration varies between 16 and 18% and CO_2 between 1 and 3%, but as m. c. increases above the limit of biological activity, O_2 concentration drops to 0–5%, while CO_2 rises up to 30%. In some cases, with exceptionally wet grain, CO_2 concentration can reach values as high as 70% (Bartosik et al., 2008b). Nevertheless, even in dry grain storage a substantial change in the internal atmosphere can be observed. This normally occurs due to the high biological activity (respiration) of a small amount of spoiling grain caused by the entry of rainwater through a perforation (Bartosik et al., 2008a).

3. Relation between a_w , airtightness, microbial activity and mycotoxins

The particular microbiota developed during storage of grains in silo-bags was investigated mainly for corn (Castellari et al., 2010, 2015; Pacin et al., 2009) and wheat (Darby and Caddick, 2007; Gregori et al., 2013). In these studies, the predominant fungal genera were *Aspergillus*, *Penicillium*, *Fusarium* and *Eurotium*.

It is generally accepted that when grain is stored at m. c. equivalent to $a_w \leq 0.7$, mold activity is prevented (Table 1). Above that threshold, molds are active and temperature plays a key role in its regulation and dominance (Marín et al., 1998). On the other hand, molds are generally

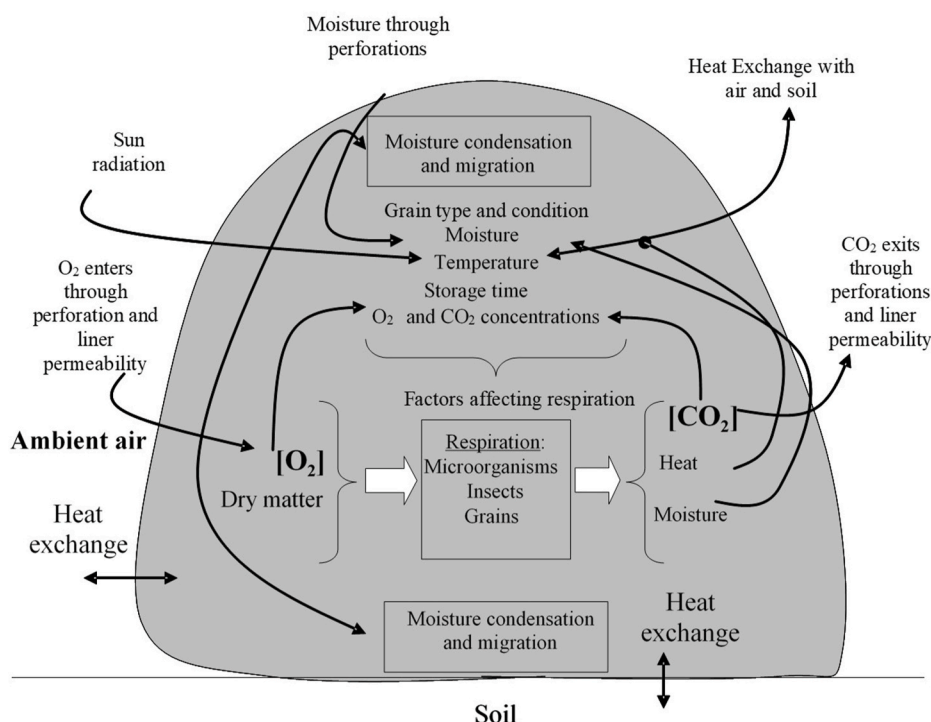


Fig. 2. Schematic representation of the biotic and abiotic components of the silo-bag ecosystem and their interactions. Adapted from Bartosik (2012).

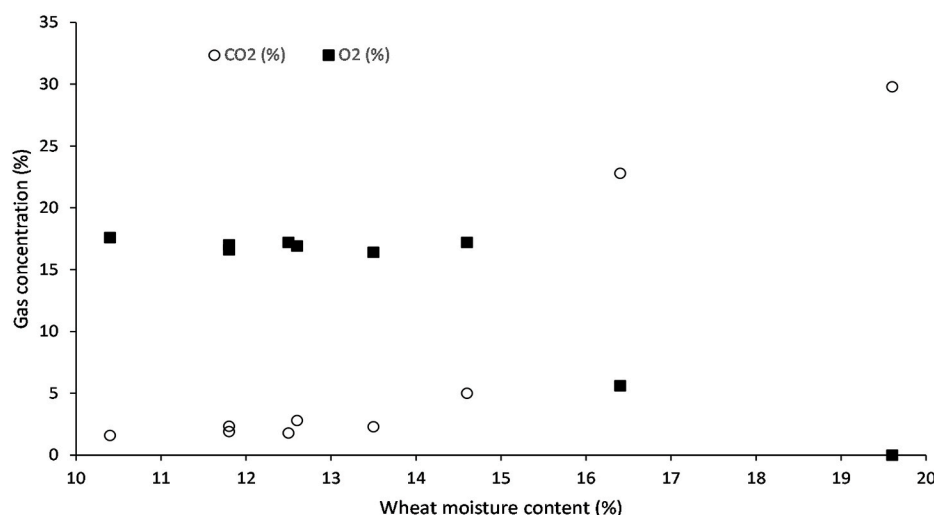


Fig. 3. O₂ and CO₂ concentration during storage of wheat with different moisture contents in different silo-bags. Adapted from Rodríguez et al. (2008).

Table 1

Moisture content equivalent to a_w of 0.7 (onset of microbial activity) and 0.84 (onset of mycotoxin production) at 20 °C for different grains. Values obtained with the INTA-ISU Grain Aeration and Storage App (Maier and Bartosik, 2020) using the equilibrium relative humidity values of 70 and 84% and the moisture relationship equations and parameters from ASAE (2001).

Grain	m.c. (%) equivalent to $a_w = 0.7$	m.c. (%) equivalent to $a_w = 0.84$
Wheat	14.9	18.1
Barley	15.3	18.6
Corn	14.0	16.6
Sunflower	8.3	12.5
Sorghum	15.9	18.6
Soybean	12.8	16.5
Canola	9.2	13.8
Rice	14.6	17.4

considered obligate aerobes, but the O₂ partial pressure they require to grow is often overestimated, as some species can grow with O₂ concentrations even below 3% (Magan and Lacey, 1984). Modified atmospheres are characterized by the formation of hypoxic or anoxic atmospheres that cause a self-regulation effect of the biological activity (Navarro et al., 2012). Microbial activity during airtight storage is under the effect of a triple interaction among a_w , O₂ and temperature. Marcos Valle et al. (2021) studied these interactions for corn and proposed a general model to explain the microbial evolution during hermetic storage. (Fig. 4). For a_w below 0.70 (dry grain), moisture is not available for microorganisms, biological activity is scarce (respiration in meaningful), and the internal atmosphere in the silo-bag remains about the same as in ambient atmosphere (Fig. 3). As a_w increases above 0.70, moisture becomes available for microbial activity, respiration rate increases and so does the modification of the internal atmosphere of the silo-bag. The aerobic phase is characterized by the activity of filamentous fungi, which tends to decrease as O₂ concentration approaches zero. If a_w is lower than 0.85, after O₂ is depleted there is a transition into a stable anaerobic phase, which is characterized by a severe limitation of microbial activity due to the effect of hermetic storage. When a_w is greater than 0.85, a transition to an active anaerobic phase is observed as O₂ is depleted. This transition is characterized by the activity of facultative anaerobic microorganisms, such as yeast, and, if a_w is greater than 0.87–0.90, by bacteria. In the active anaerobic phase, microbial activity continues through fermentation processes. The active anaerobic phase is more intense as a_w approaches 1, resulting in an exponential increase of CO₂ concentration. Finally, there is a transition from the active to the stable anaerobic phase. The mechanism that allows this transition could

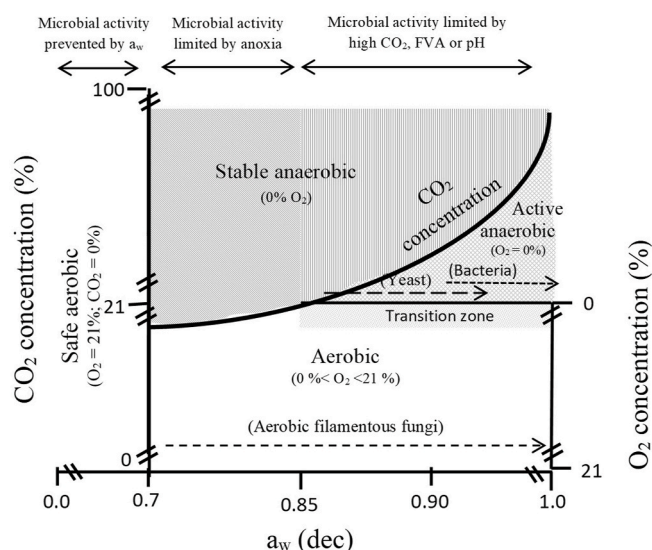


Fig. 4. Conceptual diagram of the phases of hermetic storage according to the a_w of the environment. From Marcos Valle et al. (2021a)

be related to the buildup of CO₂ concentration or the production of volatile fatty acids (Diawara et al., 1986; Magan and Lacey, 1984; Weinberg et al., 2008).

A few works have studied the presence of mycotoxins in grain stored in silo-bags (Alaniz Zanon et al., 2022; Behr et al., 2021; Castellari et al., 2015; Gregori et al., 2013; Pacin et al., 2009; Subramanyam et al., 2012). In general, these works report concentration values of different toxins at the beginning and at the end of the storage period, but they do not delve into the underlying mechanisms of microbiological activity and toxin formation in the particular environment of the silo-bag. Fleurat-Lessard (2017) summarized a_w limit values for toxin production in stored grains for different mold species and found the lowest limit to be 0.84 for *A. flavus*. Table 1 shows the equivalent m. c. for different grains. Storing grains above those thresholds of m. c. increases the risk of mycotoxin formation. On the other hand, modified atmospheres can prevent the development of mycotoxins if the O₂ concentration drops to less than 5% (Paster and Bullerman, 1988) or CO₂ concentration increases above 25% (Giorni et al., 2008). However, due to gas leakage from silo-bag, wet grain may not always achieve a critical modification of the internal atmosphere to prevent mycotoxins formation (less than

5% O₂ or more than 25% CO₂), pointing out the importance of achieving and maintaining a high level of airtightness of the silo-bag during storage to minimize the risk of toxin formation.

As mentioned in previous sections, the variables that determine the biological activity of the bulk are dynamic. The average grain temperature during storage in a silo-bag fluctuates throughout the seasons, the airtightness can be lost due to perforations in the plastic cover increasing the exchange of gases between the inside and the outside of the silo-bag (and changing the gas composition), and a_w can increase locally due to moisture migration or rainwater ingress through perforations. As a result, mold activity may not be constant during storage. Therefore, knowledge of the effect of these relationships on biological activity and their evolution, and monitoring of critical variables are important for proper quality management and mycotoxin risk assessment of grains stored in silo-bags.

4. Insects

a. Prevalence of insects in silo-bags

To our knowledge, Massigoe et al. (2010) is the only work that documented the presence of insects in silo-bags. In their barley storage study, the authors reported the presence of live insects in 6 bags of 56 evaluated. Other than that, there are some anecdotal reports of insects in silo-bags. For instance, the wheat milling industry in Argentina uses silo-bags to store dry wheat and indicates that the presence of insects is rare, but more likely during the summer and in silo-bags with grain coming from a previous storage in standard metal bins (not recently harvested). Darby and Caddick (2007) mentioned similar anecdotal reports for dry barley stored under Australian conditions. Coincidentally, Taher et al. (2014) surveyed silo-bag users and did not identify the presence of insects, or their damage to grain, as a problem.

The reasons for the low insect activity reported in silo-bags could be understood by analyzing the three grain infestation pathways: 1) most of the silo-bags are filled with grain coming directly from the harvest. The presence of insects in the field depends on environmental conditions at harvest time (temperature, r. h.), crop conditions (grain moisture, neighboring crops, etc.), and proximity to storage structures (Adler et al., 2022; Howlader and Matin, 1988; Trematerra, 2015). Consequently, grain is usually harvested free of insects (or with an unnoticeable infestation level), at least under typical Argentine production conditions; 2) each silo-bag is new (not reused), so they come free of insects, in contrast to standard metal or concrete bin that can have an infestation prior to harvest (Abadía and Bartosik, 2013; Hagstrum et al., 2012). Furthermore, during the harvesting operation and subsequent storage in silo-bags, the grain has little chance of being colonized by insects, as there is less handling compared to storage in standard grain elevators (receiving pits, bucket elevators, conveyors, etc); 3) once the grain is stored in the silo-bag, the plastic cover acts as a physical barrier, preventing the entrance of insects, contrary to what happens in conventional storage where stored grain is usually colonized by migrating insects (Hagstrum, 2001).

b. Effect of internal environment and self-modified atmosphere on insects

In cases where stored grain is colonized by insects, the development of their population will be affected by the particular environment in the silo-bag. The temperature of the grain inside the silo-bag follows the average ambient temperature throughout the year. During the fall and winter - in temperate and cold climates - grain temperature drops below the range of insect activity (15–17 °C) (Fig. 1) (Fields, 1992). The internal atmosphere also plays a role. It is generally accepted that to achieve total insect control the O₂ concentration should drop below 2% or the CO₂ concentration should increase above 20% (Banks et al., 1990; Banks and Annis, 1990; Navarro et al., 2012). Thus, when grain is stored

wet enough to support mold activity (Table 1), the modification of the internal atmosphere could be lethal for stored insects (Fig. 3). However, many times the internal atmosphere stabilizes at intermediate concentrations that might not be lethal for insects, but might still affect population growth. Subramanyam et al. (2012) conducted trials under field conditions in Kansas (USA) with silo-bags filled with hard red winter wheat, in which were inserted vials with adults of *R. dominica* (F.). They observed that the insect population inside the bag increased from 14 to 19 times lower when compared to the control (growth in chamber at 28 °C and 65% r. h.). This limitation in population growth could be related to the restricting temperature, intermediate O₂ and CO₂ concentrations generated, or a combination of both.

Insect development in the silo-bags (and damage in grain) is favored when the bag is filled with previously infested grain, grain is stored over summer time (or with grain temperature between 25 and 30 °C), and when grain is dry enough to prevent the formation of a lethal atmosphere for insects. However, even under this condition the outbreak of an insect population would be prevented by the effect of hermetic storage. Navarro et al. (1994) studied the interdependent changes in gas concentrations and dynamics of insect populations under hermetic storage, establishing that the oxygen ingress rate should be restricted to 0.05%/day to arrest insect population growth. Considering the dimensions of a standard silo-bag (60 m long and 2.76 m diameter, holding 200 t of grain), the oxygen transmission rate (OTR) of the plastic liner should be of 162 ml/(m² day) to limit the oxygen ingress rate to 0.05%/day – calculations not shown –, in the same order than that derived from reported permeance values for silo-bag liner (10.34 10⁻⁴ m³d⁻¹m⁻²atm⁻¹ – by Chelladurai et al. (2016a)). Arias Barreto et al. (2016) coupled a population growth rate model for two insect species with a silo-bag heat and mass transfer model and simulated wheat storage conditions for different regions of Argentina. They concluded that insect population growth would be limited by temperature in the Central and Southern regions of Argentina, but not in the sub-tropical regions in the north of the country. Additionally, they found that for an initial density of 1 insect/kg, the O₂ concentration ranged from 2 to 5% V/V, arresting insect development since the O₂ ingress rate remained below 0.05% per day.

c. Control of insect in silo-bags

In the previous section it was established that the particular conditions of the silo-bag do not favor the development of insects and that, although insects may be present, there are no reports of severe damage in grain. However, due to regulations that require grains to be free of live insects for marketing (zero tolerance for live insects), their control might still be necessary. In this case, the use of gaseous treatments seems to be the best control option, taking advantage of the tightness of the silo-bag (Cardoso et al., 2012). There are few works reporting successful phosphine fumigation trials in silo-bags with a dosage between 1 and 2 g/T of phosphine (Cardoso et al. (2009); Ridley et al. (2011)). Additionally, Carpaneto et al. (2016) documented the implementation of CA treatments injecting CO₂ in a full scale silo-bag to reach an initial concentration of 80%, which remained above 60% after 24 days.

5. Effect on quality

Quality evolution during storage has been evaluated for several grains including corn (13 references), wheat (6), barley (4), soybean (4), canola (4), beans (4), rice (2), sunflower (1), peanut (1), cotton seeds (1), sorghum (1), sesame seeds (1), and by-products such as dry distilled grains with solutes (DDGS) (1) and sunflower pellets (1) (Table 2). Most of these studies lasted between 6 and 9 months and were conducted either in real scale at field condition or in temperature controlled chambers using small bags (made of the standard silo-bag liner of 230 µm thickness). In general, the explanatory variables considered were m. c., temperature and CO₂ concentration. In fewer studies, equilibrium r.

Table 2

Published studies of different grains stored in full-scale silo-bags or in small bags, explanatory variables (humidity, temperature and atmosphere), reported quality parameters, country where the study was carried out and related climate characteristics according to the Köppen-Geiger climate classification (Beck et al., 2018).

Product	Explanatory variables			Test type	Evaluation parameters	Country/Climate	Source
	Hum	T	Atm				
Wt-Co-Sb-Sn	m.c.	*	CO ₂	F	TW, DK, FM, GT, PC, OA	Ar/Cfa-Cfb	Bartosik et al. (2008b)
Wt-Ba-Ca	m.c.	*	O ₂ /CO ₂	F	GT, BQ, In	Au/Cfa-Cfb-Csa-Csb	Darby and Caddick (2007)
Wt-Co	a _w	*	O ₂ /CO ₂	F	FI, CFU, My	It/Cfb-Csa	Gregori et al. (2013)
Co-Sb	m.c.	–	–	F	BK, TKW	Ar/Cfa-Cfb	Cardoso et al. (2007)
Wt	m.c.	*	O ₂ /CO ₂	F	TW, GT, BQ	Ar/Cfa-Cfb	Rodríguez et al. (2008)
Wt	m.c./r.h.	*	CO ₂	F	TW, P%, FN, TKW, CFU, My, In	USA/Cfa-BSk	Subramanyam et al. (2012)
Wt	m.c.	*	–	F	CFU, PC, pH, GT	Ge/Cfb	Idler et al. (2012)
Ba	m.c.	*	CO ₂	F	GT	Ar/Cfa-Cfb	Bartosik et al. (2012)
Ba	m.c.	*	CO ₂	F	P%, GT	Ar/Cfa-Cfb	Cardoso et al. (2010)
Ba	m.c.	*	CO ₂	F	P%, GT, PQ, In	Ar/Cfa-Cfb	Massigoe et al. (2010)
Co	m.c.	*	O ₂ /CO ₂	F	FI, CFU, My, pH	Ar/Cfa-Cfb	Castellari et al. (2015)
Co	m.c.	*	–	F	TW, DK, BK, FM, PC, My	Ar/Cfa-Cfb	Azcona et al. (2014)
Co	m.c.	*	O ₂ /CO ₂	F	In	Ar/Cfa-Cfb	Casini and Santa Juliana (2009)
Co	m.c.	*	–	F	FI, CFU	Ar/Cfa-Cfb	Castellari et al. (2010)
Co	m.c.	–	–	F	PC, GT	Br/Aw	Ruffato et al. (2014)
Co	m.c.	*	–	F	TW, TKW, GT	Br/Aw	Taffarel et al. (2013)
Co	m.c.	*	CO ₂	F	TW, DK, In.	Mx/Csb	Recendiz Hurtado et al. (2011)
Co	m.c.	–	–	F	TW, DK, BK, FM, FI, My	Ar/Cfa-Cfb	Díaz et al. (2012)
Co	m.c./a _w	–	–	F	CFU, FI, My	Ar/Cfa-Cfb	Pacin et al. (2009)
Co	m.c./r.h.	*	O ₂ /CO ₂	S	TW, DK, CFU, GT, In, My	Ar/Cfa-Cfb	Cardoso et al. (2016)
CS	m.c.	–	CO ₂	F	PC	Ar/Cfa-Cfb	Hidalgo et al. (2014)
Sb	m.c./r.h.	*	–	F	GT	Ar/Cfa-Cfb	Clemente et al. (2009)
Sb	m.c.	–	CO ₂	F	DK	Ar/Cfa-Cfb	Taher et al. (2019)
Ri	m.c.	*	–	F	BK, DK, GT	Ar/Cfa-Cfb	Cardoso et al. (2014)
Ri	m.c.	–	–	F	BK	Ar/Cfa-Cfb	Hidalgo et al. (2012)
Ca	m.c.	*	CO ₂	F	GT, FAV	Ca/Dfb	Chelladurai et al. (2016; 2016b)
Ca	m.c./r.h.	*	CO ₂	F	O%, OAI	Ar/Cfa-Cfb	Ochandio et al. (2010)
Pe	m.c./r.h.	*	O ₂	F	PQ	USA/Cfa	Butts and Ward (2022)
Sn pellets	m.c./r.h.	*	CO ₂	F	PC, PQ, CFU, My	Ar/Cfa-Cfb	Behr et al. (2021)
Be	m.c./r.h.	*	CO ₂	F	BK, DK, FM	Ar/Cwa	Godoy (2014)
Be	m.c./r.h.	*	CO ₂	F	BK, DK, FM, CFU	Ar/Cfa-Cfb	Godoy (2019)
Be	m.c.	*	–	S	In, EC, CT, TW, GT	25 °C/70	Freitas et al. (2011, 2016)
Sg	m.c./r.h.	*	CO ₂	F	–	Ar/Cfa-Cfb	Ochandio et al., 2014b
WDGS	m.c.	*	O ₂ /CO ₂	S	SE, CFU, My, N%, RDP, IUPD, pH	22 °C/60–65% r.h.	Alvarez et al. (2019)
SS	m.c.	*	–	S	PC	30 °C	Marrugo-Ligardo et al. (2015)

References. Product: Ba is barley; Be is beans; Ca is canola; Co is corn; CS is cotton seeds; Pe is peanuts; Ri is rice, Sb is soybean; Sg is sorghum; Sn is sunflower; SS is sesame seeds; WDGS is wet distilled grains with solutes; Wt is wheat. Test type: F is field conditions; S is small bags in temperature controlled chambers. Evaluation parameters: BK is broken kernels; BQ is baking quality; CFU is colony forming units; CT is cooking time; DK is damaged kernels; EC is electrical conductivity; FAV is free fatty acid values; FI is fungal identification; FM is foreign matter; FN is falling number; GT is germination test; In is insects; IUPD is intestinal undegradable protein digestibility; My is mycotoxins; N% is nitrogen concentration; O% is oil concentration; OA is oil acidity index; P% is protein concentration; PC is proximate composition; PQ is physical quality parameters; RDP is ruminal degradability protein; SE is sensory evaluation; TKW is thousand kernels weight; TW is test weight. Countries: Ar is Argentina; Au is Australia; Br is Brazil; Ca is Canada; Ge is Germany; It is Italy; Mx is Mexico; USA is United States of America. Köppen-Geiger Main Climate: C is warm temperate; B is arid; A is equatorial; D is snow. Precipitation: f is fully humid; s is summer dry; w is winter dry. Temperature: a is hot summer; b is warm summer; k is cold arid.

h. (or a_w) and O₂ concentration were also measured. The quality evaluation parameters most frequently used were germination test, microbial load (quantified as colony forming units (CFU)), fungal identification and test weight (volumetric density). In addition to this, particular quality parameters were also evaluated for specific products. For instance, oil concentration, free fatty acids and oil acidity index were reported for oilseeds, baking quality for wheat, and physical properties for rice and sunflower pellets (Table 2).

The quality of the stored grains in silo-bags is affected by the interaction between m. c. and temperature. When m. c. is low enough to inhibit microbiological activity (Table 1), the temperature itself does not have major influence and the products could be stored even during the summer without quality deterioration (with some exceptions considered below) (Bartosik et al., 2008b; Cardoso et al., 2010). On the other hand, when the m. c. is high enough to support microbial activity, temperature during storage results in a major concern. It was previously mentioned that the grain temperature in silo-bags mimic the average ambient temperature throughout the year (Fig. 1). As a consequence, during winter months, quality parameters are preserved (or at least deterioration is slowed down) because the biological activity is arrested by the combination of low temperature and the modified atmosphere.

However, when storage is prolonged towards the warm seasons (spring and summer) microbial activity and other deleterious processes (eg. biochemical reactions) increase causing a quick loss in quality parameters that cannot be compensated by the modified atmosphere (Bartosik et al., 2002, 2012).

It is important to remark that most of the reported studies were carried out under warm climates, high humid, and hot or warm summer conditions (according to the Köppen-Geiger climate classification - Beck et al. (2018)), such as the Argentina Pampa's region (28 references), some under colder conditions (eg. Canada) (3 references) and only 2 works were carried out under tropical or subtropical conditions (eg. Brazil). Thus, storage recommendations should not be directly extrapolated from studies carried out in warm-temperate regions to tropical or subtropical regions without proper considerations.

There is a general consensus that the external grain layer might have a differential quality evolution that the grain at the center of the bag. In section 2. b it was stated that moisture migration to the top grain layer may increase the risk of spoilage (and quality deterioration) via microbiological activity. Along the same line, storing slightly damp grains can result in damage at the top grain layer due to biological activity fostered by the availability of moisture and high temperature

(Chelladurai et al., 2016; Jian et al., 2015a). In addition to that, Darby and Caddick (2007) raised some concerns about quality loss of malting barley in the silo-bag outermost layer due to temperature. They speculated that the external grain layer reaches such high temperature during the daytime that seed viability could be negatively affected by heat damage. The cross-sectional plane of the silo-bag can be modeled as a half-round. For this geometry, each layer of grains 0.10 m deep represents approximately 5% of the total volume. Thus, considering that the effect of the daily temperature oscillation was observed in the first 10–20 cm (Bartosik et al., 2008b; Gastón et al., 2009), the affected amount of grain for this phenomenon would be in the order of 5–10%. However, in the present review, no conclusive evidence of this damage mechanism was found for storing dry grains (m.c. below the range of biological activity). For instance Cardoso et al. (2010) collected grain samples segregated by stratus (upper, middle and lower) and did not find differences in germination tests. Nevertheless, some quality parameters that are particularly sensitive to heat damage, such as seed vigor (measured as radicle emergence (Matthews et al., 2012)), could accelerate their deterioration kinetics when they are exposed to temperature cycles with peaks above 40 °C.

6. Modelling

Two main silo-bag modeling works have been identified. The first model was developed by Gastón et al. (2009) as a bi-dimensional finite element model that predicts temperature distribution and moisture migration of wheat stored in silo-bags due to seasonal variation of climatic conditions. The model includes grain respiration and calculates CO₂ and O₂ concentrations during storage as well as the associated d.m.I. Later, Abalone et al. (2011b) applied the former model in combination with a lumped-capacity differential equation for gas transfer to simulate the gas exchange, and Abalone et al. (2011a) evaluated the effect of respiration rate, permeance and perforations on the internal atmosphere. This model was then used to study storage conditions for different climatic regions of Argentina for wheat (Arias Barreto et al., 2013b) and soybean (Arias Barreto et al., 2017). Finally, Arias Barreto et al. (2016) incorporated sub-models of the population growth rate of *R. dominica* (Fabricius) and *S. oryzae* (L.) and studied their dynamics under different storage conditions. The second main model was developed by Jian et al. (2015b) and combined soil models based on the energy balance on the surface of the ground and heat conduction equations, with a three-dimensional transient heat, mass, and momentum transfer models inside silo-bags. This model was used to predict temperatures, moisture contents and condensation of canola stored inside silo-bags under Canadian Prairie conditions (Jian et al., 2015a).

7. Airtightness protection

The main causes of loss of airtightness are unsuitable soil preparation, inappropriate sealing practice, and fauna activity (Cardoso et al., 2012; Taher et al., 2014). The best management practices for achieving and maintaining a suitable airtightness were summarized by Abadía and Bartosik (2013a), and include the adequate preparation of the soil to remove stones, hard stems of crops or weeds that can perforate the bag from below, heat sealing of the ends of the bag, and preventive measures against wildlife and domestic animal activity. Zufiaurre et al. (2020, 2018) determined that armadillos (*Chaetophractus villosus*), rodents and dogs (in that order) were the most harmful species. The recommendation for minimizing vertebrate damage includes keeping the grass short, setting up a fence around the bags and spraying natural repellents (Abadía and Bartosik, 2013).

8. Monitoring

a. Pressure decay test for airtightness assessment

Assessing the physical integrity of a silo-bag in the field is rather difficult. Small perforations are hard to spot, particularly if they are located in the top or bottom of the bag. A practical methodology for evaluating the airtightness of any hermetic system is through a pressure decay test (PDT). Navarro and Zettler (2000) correlated different levels of perforated areas with the leakage of gasses and concluded that for flexible structures of less than 500 t of capacity and full of grain a PDT of 1.5 min, 3 min and 5 min were required for fumigation (phosphine), controlled atmosphere and modified atmosphere treatments, respectively. With the intention of making a quick assessment of the silo bag airtightness, Bartosik et al. (2016) developed an automatic pressure decay test device, consisting of a portable unit with electrical fans, connecting hoses, pressure gauge, valves and Bluetooth connectivity to a cell phone.

b. Biological activity and grain quality losses

Specific monitoring strategies were also developed for silo-bag storage. The degree of modification of the silo-bag internal gas composition can be related to the activity of biotic components of the grain bulk (Cardoso et al., 2008; Rodríguez et al., 2008) and used for early detection of spoilage (Bartosik et al., 2008a). A specific monitoring device was developed based on this concept (Silcheck, Lincoln, Argentina) in which each silo-bag is tagged with a radio frequency identification (RFID) card and the internal CO₂ concentration measured every 6 m along the bag with a portable meter. The information is transferred to a server and analyzed with a proprietary algorithm to assess the storage risk (Bartosik et al., 2013). Later, Taher et al. (2019) developed a prediction model for soybean losses (kg of spoiled grain not commercialized) in silo-bag storage based on monitoring the CO₂ concentration and other sensible variables.

9. Economics and logistics

A survey conducted by Taher et al. (2019) showed that the use of the silo-bag allow users to achieve savings in lower freight rates, broker's commercial commission or to obtain a small premium on the selling price – all factors having a positive impact of the economic margins. The authors concluded that the low level of recorded losses and the possibility to obtain higher economic margin explain in part the widespread adoption of this technology. These results agree with those of Darby and Caddick (2007) who interviewed Australian farmers using the silo-bag technology. The authors found that this technology better positioned its users to negotiate grain prices, broker commissions or grain handling charges with grain buyers which increased their economic margins.

Urcola and Bartosik (2020) developed a bio-economic model of soybean storage considering the risk of grain losses. The model compares the economic outcomes of storing soybeans in silo-bags, as compared to using a commercial storage facility, and allows for adapting the length of storage to minimize grain losses. Results indicated that, if no losses occur, storing soybeans in the silo-bag yielded profits 1.7% higher than storing it in a commercial storage facility. Profits from the silo-bag and from the grain storage facility break-even at 2% soybean losses. When storing in the silo-bag results in the loss of 5% of the grain, the silo-bag yielded profits 2.1% lower than the commercial storage facility. This study shows that when best management practices are employed and grain losses are kept to a minimum, the silo-bag constitutes a safe and profitable technology for grain storage.

The review of the existing literature suggests that there are several productive and logistic advantages of using the silo-bag (Moreno Ferro and Paturanne, 2015; Taher et al., 2019). Nonetheless, most studies attempting to estimate the value of this technology focused on financial benefits, with the exception of Busato et al. (2011) who implemented a discrete event simulation to model the impact of silo-bags on the harvest and transportation logistics at the farm level. They compared the scenarios of harvesting and storing grain in a silo-bag on the farm versus

harvesting and sending the production to a commercial grain elevator. The results showed that the greater the distance from the farm to the elevator, the greater the reduction in costs due to the use of silo-bags. In particular, a notable increase in the efficient use of combines could be verified, since silo-bags avoid the typical interruptions due to delays in transportation or the operation of grain elevators at the peak of harvest were avoided.

At the country level, Gatti (2015) estimated the social benefit of the silo-bags for the Argentine agricultural sector, considering the supply and demand for grain storage services. Quantifying the surplus obtained by producers of storage services plus the one obtained by the consumers of these services, the authors estimated that this innovation generated an aggregate benefit ranging between 8.5 and 10.5 U.S. bill dollars. Additionally, the adoption of silo-bags has created the conditions to promote the development of several businesses associated with this technology, generating a significant amount of income for the country.

10. Future research needs

Based on the current review of the state of the art of the silo-bag storage system, the following issues were detected as gaps in knowledge that need to be addressed in future research projects:

In relation to storage conditions and effect on product quality it is necessary to deepen the understanding of the effect of critical variables (temperature, gases and equilibrium relative humidity) on grain quality for tropical climates; study the cyclical effect of temperature (day/night) and consequent moisture migration in grain quality losses in the outermost silo-bag layer; and better understand the mechanisms of mycotoxin formation in this particular storage system. On the other hand, rice is one of the most important staple foods in the world, and yet there are very few studies on the storage of rice in silo-bags. Given the importance of this crop, the particular conditions in which it is produced (mainly tropical and subtropical regions of the world), harvested (high moisture content) and stored (with husk), it is necessary to carry out studies about the storage of rice in silos-bag for its most representative conditions.

It is necessary to document the incidence of insects in silo-bags and better understand the effect of the particular silo-bag environment on insect population dynamics and subsequent grain damage. Additionally, given the growing demand for treatments free of pesticide residues and for organic products, the silo-bag technology could play an important role in gas-based pest control schemes in the future. Hence, it is also required to further evaluate the use of silo-bags for implementing economical and effective fumigation and controlled atmosphere treatments under field conditions.

In terms of modeling it is necessary incorporate the effect of gas diffusion from localized hotspots (eg. caused by rainwater ingress through perforations in the plastic cover) inside the silo-bag in order to refine monitoring tools based on CO₂ measurement. Modeling the effect of critical storage variables on grain quality parameters, such as seed viability, mycotoxins production, baking quality in wheat, etc, could help better understand quality kinetics during storage under different conditions. This would help to improve and refine recommendations about safe storage conditions in silo-bags. Additionally, existing simulation models, with the enhancements described above, could be used to carry out preliminary evaluations about the effect of silo-bag storage on grain quality in different climatic regions and agricultural systems, where field studies are not yet available. This would allow anticipating problems, such as moisture migration and condensation, loss of viability in seeds in the outer layer of the bag, etc.

In relation to the impact of the use of silo-bags on the logistics and economic aspects of agricultural activity, several issues that need to be addressed were identified. At the farm level, it is necessary to deepen the quantification of the impact of the use of silo-bags on logistics, such as greater harvest speed, operational flexibility, segregation, identity preservation, traceability, etc., and the economic benefits it provides. At

the regional or country level, it is necessary to study the impact of the use of silo-bags on the logistics and economy of the entire agricultural sector. This information would be critical for making decisions on the incorporation of this storage technology in other countries or regions, where the silo-bag could help fill the lack of storage capacity and reduce post-harvest losses.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abadía, B., Bartosik, R., 2013. *Manual de Buenas Prácticas en Poscosecha de Granos*. [Grain postharvest best management practices handbook]. INTA Ediciones, Buenos Aires, Argentina. Primera.
- Abalone, R., Gastón, A., Bartosik, R., Cardoso, L., Rodríguez, J., 2011a. Gas concentration in the interstitial atmosphere of a wheat silo-bag. Part I: model development and validation. *J. Stored Prod. Res.* 47, 268–275. <https://doi.org/10.1016/j.jspr.2011.05.004>.
- Abalone, R., Gastón, A., Bartosik, R., Cardoso, L., Rodríguez, J., 2011b. Gas concentration in the interstitial atmosphere of a wheat silo-bag. Part II: model sensitivity and effect of grain storage conditions. *J. Stored Prod. Res.* 47, 276–283. <https://doi.org/10.1016/j.jspr.2011.05.003>.
- Adler, C., Athanassiou, C., Carvalho, M.O., Emekci, M., Gvozdenac, S., Hamel, D., Riudavets, J., Stejskal, V., Trdan, S., Trematerra, P., 2022. Changes in the distribution and pest risk of stored product insects in Europe due to global warming: need for pan-European pest monitoring and improved food-safety. *J. Stored Prod. Res.* 97, 101977 <https://doi.org/10.1016/J.JSPR.2022.101977>.
- Alvarez, E., Cardoso, M., Depetris, G., Castellari, C., Cristos, D., Montiel, M.D., Bartosik, R., 2019. Storage of WDGs under hermetic and non-hermetic conditions: Effect on sensory properties, microorganisms, mycotoxins and nutritional value. *J. Stored Prod. Res.* 80, 65–70. <https://doi.org/10.1016/J.JSPR.2018.11.011>.
- Al-Yahya, S.A., Bern, C.J., Misra, M.K., Bailey, T.B., 1993. Carbon dioxide evolution of fungicide-treated high-moisture corn. *Trans. ASAE (Am. Soc. Agric. Eng.)* 36, 1417–1422.
- Alaniz Zanon, M.S., Bossa, M., Chiotta, M.L., Oddino, C., Giovanini, D., Cardoso, L., Bartosik, R., Chulze, S., 2022. Pre-harvest strategy for reducing aflatoxin accumulation during storage of maize in Argentina. *Int. J. Food Microbiol.* 380, 1–9. <https://doi.org/10.1016/j.ijfoodmicro.2022.109887>.
- Alvarez, E., Cardoso, L., Bartosik, R., Depetris, G., Castellari, C., Montiel, M.D., 2016. A preliminary comparative study of conventional and hermetic storage of wet distillers grains with solubles. In: Navarro, S., Jayas, D., Alagusundaram, K. (Eds.), *Proceedings of the 10th International Conference on Controlled Atmospheres and Fumigation in Stored Products*. CAF Permanent Committee, New Delhi, India, pp. 324–328. November 6–11 of 2016.
- Arias Barreto, A., Abalone, R., Gastón, A., 2016. Prediction of insect development in a wheat (*Triticum aestivum*) silo-bag by computer simulation. In: Navarro, S., Jayas, D.S., A., K. (Eds.), *Proceedings of the 10th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF2016)*. CAF Permanent Committee Secretariat, Winnipeg, Canada, pp. 316–323.
- Arias Barreto, A., Abalone, R., Gastón, A., 2013a. Mathematical modelling of momentum, heat and mass transfer in grains stored in silos. Part I: model development and validation. *Lat. Am. Appl. Res.* 43, 337–384.
- Arias Barreto, A., Abalone, R., Gastón, A., Bartosik, R., 2013b. Analysis of storage conditions of a wheat silo-bag for different weather conditions by computer simulation. *Biosyst. Eng.* 116, 497–508.
- Arias Barreto, A., Abalone, R., Ochandio, D., Cardoso, L., Bartosik, R., 2017. Validation of a heat, moisture and gas concentration transfer model for soybean (*Glycine max*) grains stored in plastic bags (silo bags). *Biosyst. Eng.* 8, 23–37. <https://doi.org/10.1016/j.biosystemseng.2017.03.009>.
- Azcona, J., Schang, M., Couterot, A., Sala, M., 2014. Calidad nutricional de maíz almacenado en silo bolsa. [Nutritional quality of corn stored in silo-bag]. In: Cardoso, L., Bartosik, R., de La Torre, D., Abadía, B., Santa Juliana, M. (Eds.), *Almacenamiento de Granos En Silo Bolsa: Resultados de Investigación 2009-2013*. Balcarce, pp. 86–93.
- Banks, H.J., Annis, P.C., 1990. Comparative advantages of high carbon dioxide and low oxygen types of controlled atmospheres for grain storage. In: *Food Preservation by Modified Atmospheres*, pp. 93–122.
- Banks, H.J., Annis, P.C., Rigby, G.R., 1990. Controlled atmosphere storage of grain: the known and the future. In: Fleurat-Lessard, F., Ducom, P. (Eds.), *Proceedings Fifth International Working Conference on Stored Product Protection*. IWCSPP, Mompelie, France, pp. 695–706.

- Bartosik, R., 2012. An inside look at the silo-bag system. In: Navarro, S., Banks, H.J., Jayas, D.S., Bell, C.H., Noyes, R.T., Ferizli, A.G., Emekci, M., Isikber, A.A., Alagusundaram, K. (Eds.), *Proceedings of the 9th International Conference Controlled Atmospheres and Fumigation of Stored Products*. CAF, Antalya, Turkey, pp. 117–128.
- Bartosik, R., Cardoso, L., Albino, J., Busato, P., 2013. CO₂ monitoring of grain stored in silobag through a web application. In: *Proceedings of the EFITA, WCCA, CIGR 2013 Conference: Sustainable Agriculture through ICT Innovation*. CIGR, Torino, Italy, pp. 24–27.
- Bartosik, R., Cardoso, L., de la Torre, D., Abadía, B., Igartua, M., 2016. An approach for developing a phosphine dosage procedure and calculator for silo bags. In: Navarro, S., Jayas, D., Alagusundaram, K. (Eds.), *Proceedings of the 10th International Conference on Controlled Atmospheres and Fumigation in Stored Products*, pp. 371–376. New Delhi, India, November 6–11 of 2016.
- Bartosik, R., Cardoso, L., Rodríguez, J., 2008a. Early detection of spoiled grain stored in hermetic plastic bags (silo-bags) using CO₂ monitoring. In: Daolin, G., Navarro, S., Jian, Y., Cheng, T., Zuxun, J., Yue, L. (Eds.), *Proceedings of the 8th International Conference Controlled Atmospheres and Fumigation of Stored Products*. Sichuan Publishing House of Science and Technology, Chengdu, China, pp. 550–554.
- Bartosik, R., Ochandio, D., Cardoso, L., de la Torre, D., 2012. Storage of malting barley with different moisture contents in hermetic silo bags. In: Navarro, S., Banks, H.J., Jayas, D.S., Bell, C.H., Noyes, R.T., Ferizli, A.G., Emekci, M., Isikber, A.A., Alagusundaram, K. (Eds.), *Proceedings of the 9th International Conference Controlled Atmospheres and Fumigation of Stored Products*, October 5–19 of 2012. CAF Permanent Committee, Antalya, Turkey.
- Bartosik, R., Rodríguez, J., Cardoso, L., 2008b. Storage of corn, wheat, soybean and sunflower in hermetic plastic bags. Chicago, Illinois, USA. In: *Proceeding of the 2008 International Grain Quality & Technology Congress*, 0–13.
- Bartosik, R., Rodríguez, J., Malinarich, H., Maier, D.E., 2002. Silobag™: evaluation of a new technique for temporary storage of wheat in the field. IWCSPP. In: *Proceedings of the 8th International Working Conference on Stored Product Protection*, pp. 1018–1023. York, England.
- Beck, H.E., Zimmermann, N.E., McVicar, T.R., Vergopolan, N., Berg, A., Wood, E.F., 2018. Present and future Köppen-Geiger climate classification maps at 1-km resolution. *Sci. Data* 5, 180214. <https://doi.org/10.1038/sdata.2018.214>.
- Behr, E., Cardoso, L., Bartosik, R., Marcos Valle, F., de la Torre, D., Taher, H., Maciel, G., 2021. Evaluation of storage of sunflower pellets in silo bags. In: Jayas DS, J.F. (Ed.), *Proceedings of the 11th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF2020)*. CAF Permanent Committee Secretariat, Winnipeg, Canada, pp. 123–130, 23–27 August 2021.
- Bern, C.J., Steele, J.L., Morey, R.V., 2002. Shelled corn CO₂ evolution and storage time for 0.5% dry matter loss. *Appl. Eng. Agric.* 18, 703–706.
- Busato, P., Berruto, R., Cardoso, L., Bartosik, R., 2011. Logistics and economics of grain harvest and transport systems with the use of silo-bag. In: *American Society of Agricultural and Biological Engineers Annual International Meeting 2011*. ASABE, pp. 5503–5512. ASABE.
- Butts, C.L., Ward, J.K., 2022. Storing peanuts in grain bags. *Appl. Eng. Agric.* 38, 93–102. <https://doi.org/10.13031/AEA.14475>.
- Cardoso, L., Bartosik, R., Campabadal, C., de la Torre, D., 2012. Air-tightness level in hermetic plastic bags (silo-bags) for different storage conditions. In: Navarro, S., Banks, H.J., Jayas, D.S., Bell, C.H., Noyes, R.T., Ferizli, A.G., Emekci, M., Isikber, A.A., A.K. (Eds.), *Proc 9th. Int. Conf. On Controlled Atmosphere and Fumigation in Stored Products*. CAF, Antalya, Turkey - October 15–19, pp. 583–589.
- Cardoso, L., Bartosik, R., de la Torre, D., Abadía, B., Santa Juliana, M., 2014. Almacenamiento de granos en silo bolsa. Resultados de investigación 2009–2013. Ediciones INTA, Buenos Aires, Argentina [Storage of grains in silo-bag. Research report 2009–2013].
- Cardoso, L., Bartosik, R., Milanese, D., 2009. Phosphine concentration change during fumigation in hermetic plastic bags (Silobags). In: *Proceedings of the CIGR Section V International Symposium*. Rosario, Argentina. CIGR.
- Cardoso, L., Bartosik, R., Rodríguez, J., 2007. Estudio de la evolución de la humedad de los granos individuales en silo bolsas de maíz y soja. [Study of the individual kernel moisture content of corn and soybean stored in silo bags]. In: *Memorias Del IX Congreso Argentino de Ingeniería Rural y I Del MERCOSUR*. Universidad Nacional de Córdoba, Ciudad de Córdoba, Córdoba, Argentina, pp. 1–22. September 19–22.
- Cardoso, L., Bartosik, R., Rodríguez, J., Ochandio, D., 2008. Factors affecting carbon dioxide concentration in interstitial air of soybean stored in hermetic plastic bags (silo-bag). In: Daolin, G., Navarro, S., Jian, Y., Cheng, T., Zuxun, J., Yue, L. (Eds.), *Proceedings of the 8th International Conference Controlled Atmospheres and Fumigation of Stored Products*. Sichuan Publishing House of Science and Technology, Chengdu, China, pp. 565–568.
- Cardoso, L., Ochandio, D., de la Torre, D., Bartosik, R., Rodríguez, J., 2010. Storage of quality malting barley in hermetic plastic bags. In: *International Working Conference on Stored Product Protection*, pp. 331–337. <https://doi.org/10.5073/jka.2010.425.167.322>, 10th International Working Conference on Stored Product Protection. Julius-Kühn-Archiv, Estoril, Portugal. June 27 - July 2 of 2010.
- Carpaneto, B., Bartosik, R., Cardoso, L., Manetti, P., 2016. Pest control treatments with phosphine and controlled atmospheres in silo bags with different airtightness conditions. *J. Stored Prod. Res.* Submitted 143–151. <https://doi.org/10.1016/j.jspr.2016.07.007>.
- Cardoso, L., Bartosik, R., Castellari, C., Abadía, B., de la Torre, D., Taher, H., 2016. Hermetic storage of wet corn in liners with and without EVOH barrier. In: Navarro, S., Jayas, D., Alagusundaram, K. (Eds.), *Proceedings of the 10th International Conference of Controlled Atmospheres and Fumigation of Stored Products*. CAF Permanent Committee, New Delhi, India, pp. 117–128. November 6–11 of 2016.
- Casini, C., 2014. Technological evolution of grain storage in plastic bags. In: Gastón, A., Abalone, R., Bartosik, R. (Eds.), *1st International Conference of Silo Bag*. Instituto Nacional de Tecnología Agropecuaria (INTA), Mar del Plata/Balcarce, Argentina, p. 19. October 13 - 16.
- Casini, C., Santa Juliana, D., 2009. Determinación del efecto de la humedad del maíz sobre la concentración de O₂ y CO₂ en bolsas plásticas. [Effect of corn moisture content in concentration of O₂ and CO₂ during silo-bag storage]. In: Casini, C., Rodríguez, J., Bartosik, R. (Eds.), *Almacenamiento de Granos En Bolsas Plásticas. Resultados de Investigación*. INTA, Manfredi, Argentina, p. 180.
- Castellari, C., Cendoya, M.G., Marcos Valle, F., Barrera, V., Pacin, A.M., 2015. Factores extrínsecos e intrínsecos asociados a poblaciones fúngicas micotoxigénicas de granos de maíz (*Zea mays* L.) almacenados en silos bolsa en Argentina. [Extrinsic and intrinsic factors associated with mycotoxigenic fungal populations of corn stored in silo bags in Argentina. *Rev. Argent. Microbiol.* 47, 350–359. <https://doi.org/10.1016/j.ram.2015.08.003>.
- Castellari, C., Marcos Valle, F., Mutti, J., Cardoso, L., Bartosik, R., 2010. Toxigenic fungi in corn (maize) stored in hermetic plastic bags. In: *10th International Working Conference on Stored Product Protection*. IWCSPP, Estoril, Portugal, pp. 501–504. <https://doi.org/10.5073/jka.2010.425.323>.
- Chelladurai, V., Jian, F., Jayas, D.S., White, N.D.G., 2016a. Permeability of silo bag material for carbon dioxide and oxygen. In: Navarro, S., Jayas, D.S., A., K. (Eds.), *Proceedings of the 10th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF2016)*. CAF Permanent Committee Secretariat, Winnipeg, Canada, New Delhi, India. November 6–11 of 2016.
- Chelladurai, V., Jian, F., Jayas, D.S., White, N.D.G., Fields, P.G., Manickavasagan, A., 2016b. Feasibility of storing canola at different moisture contents in silo bags under Canadian Prairie conditions. *Can. Biosyst. Eng.* 58, 309–320. <https://doi.org/10.7451/CBE.2016.58.3.9>.
- Chelladurai, V., Jian, F., Jayas, D.S., White, N.D.G., Manickavasagan, A., Fields, P.G., 2016. Quality changes in 12% moisture content canola stored in silo bags under Canadian Prairie conditions. *J. Stored Prod. Res.* 68, 33–43. <https://doi.org/10.1016/j.jspr.2016.04.002>.
- Chotikasatien, C., Chayaprasert, W., Pathaveerat, S., 2017. A study on the feasibility of quantifying the population density of stored product insects in air-tight grain storage using CO₂ concentration measurements. *J. Stored Prod. Res.* 73, 21–29. <https://doi.org/10.1016/j.jspr.2017.05.005>.
- CIDETER, 2022. Comercio exterior – fundación CIDETER [WWW document]. URL Agric. Equip. Export Rep, 8.31.22. <https://cecma.com.ar/comercio-exterior/>.
- Clemente, G., Peppi, B., Casini, C., Pagliaro, M., Mauricio, D., Juliana, S., 2009. Study of the effect of shade cloth cover on quality of soybean seeds (*Glycine max* (L.) Merr.) stored in plastic bags. In: *CIGR Proceeding of Technology and Management to Increase the Efficiency in Sustainable Ag. Systems*. CIGR, Rosario, pp. 1–4. Argentina September 2009.
- Cofie-Agblor, R., Muir, W.E., Sinha, R.N., 1995. Comparative heat of respiration of five grain beetles in stored wheat. *Postharvest Biol. Technol.* 5, 167–175.
- Darby, J., Caddick, L., 2007. Review of Grain Harvest Bag Technology under Australian Conditions.
- Diawara, B., Cahagnier, B., Richard-Molar, D., 1986. Oxygen consumption by a wet grain ecosystem in hermetic silos at various water activities. In: *Proceedings of the 4th International Working Conference on Stored-Product Protection*. Tel Aviv, Israel, September 1986, pp. 77–84.
- Díaz, C.G., Rodríguez, R., Aguayso, C., Juárez, J., Saleme, P., Ploper, L.D., 2012. Relación entre incidencia de Fusarium verticillioides y variables de calidad de grano bajo condiciones de almacenamiento de maíz en Tucumán, Argentina. [Relationship between incidence of Fusarium verticillioides and grain quality variables during maize storage in Tucumán, Argentina. *Rev. Agronómica del Noroeste Argentino* 32, 47–53.
- Faroni, L.R.D.A., de Alencar, E.R., Paes, J.L., da Costa, A.R., Roma, R.C.C., 2009. Armazenamento de soja em silos tipo bolsa. [Storage of soybean in silo-bags]. *Eng. Agric.* 29, 91–100. <https://doi.org/10.1590/S0100-69162009000100010>.
- Fields, P.G., 1992. The control of stored-product insects and mites with extreme temperatures. *J. Stored Prod. Res.* 28, 89–118. [https://doi.org/10.1016/0022-474X\(92\)90018-L](https://doi.org/10.1016/0022-474X(92)90018-L).
- Fleurat-Lessard, F., 2017. Integrated management of the risks of stored grain spoilage by seedborne fungi and contamination by storage mould mycotoxins – an update. *J. Stored Prod. Res.* 71, 22–40. <https://doi.org/10.1016/j.jspr.2016.10.002>.
- Freitas, R.S., Faroni, L.R.A., Sousa, A.H., 2016. Hermetic storage for control of common bean weevil, *Acanthoscelides obtectus* (Say). *J. Stored Prod. Res.* 66, 1–5. <https://doi.org/10.1016/j.jspr.2015.12.004>.
- Freitas, R.D.S., Faroni, L.R.D., Sousa, A.H., Cecon, P.R., Carvalho, M.S., 2011. Quality of beans stored under hermetic conditions. *Eng. Agric.* 31, 1136–1149. <https://doi.org/10.1590/S0100-69162011000600011>.
- Gaikwad, K.K., Rubino, M., 2017. Water Vapor Permeability of Packaging Materials (LDPE) and Fabricated Package Systems. East Lansing, Michigan, USA.
- Gastón, A., Abalone, R., Bartosik, R., Rodríguez, J., 2009. Mathematical modelling of heat and moisture transfer of wheat stored in plastic bags (silobags). *Biosyst. Eng.* 104, 72–85. <https://doi.org/10.1016/j.biosystemseng.2009.06.012>.
- Gastón, A., Abalone, R., Bartosik, R., Rodríguez, J., 2008. Heat and mass transfer in soybean stored in hermetic plastic bags (silobags). CIGR. In: *Proceedings of the International Conference of Agricultural Engineering, XXXVII Brazilian Congress of Agricultural Engineering*. Foss do Iguaçu, Brasil (unpaginated).
- Gatti, N., 2015. Evaluación económica de la innovación: el caso del silo bolsa. [Economic assessment of the innovation: the silo-bag case]. *Rev. Investig. Agropecu.* 41, 4.
- Giorni, P., Battilani, P., Pietri, A., Magan, N., 2008. Effect of aw and CO₂ level on *Aspergillus flavus* growth and aflatoxin production in high moisture maize post-

- harvest. *Int. J. Food Microbiol.* 122, 109–113. <https://doi.org/10.1016/J.IJFOODMICRO.2007.11.051>.
- Godoy, A., 2014. Silo bolsa de poroto. Concentración de dióxido de carbono en relación a las condiciones de almacenamiento. [Bean storage in silo bag. Carbon dioxide concentration in relation to storage conditions]. In: Cardoso, L., Bartosik, R., de la Torre, D., Abadía, B., Santa Juliana, D. (Eds.), *Almacenamiento de Granos En Silo Bolsa: Resultados de Investigación 2009-2013*. INTA Ediciones, Balcarce, Argentina, pp. 128–137.
- Godoy, A., 2019. Calidad de poroto (*Phaseolus vulgaris* L.) almacenado en silobolsa en condiciones de intemperie en relación a la presencia de materias extrañas y granos partidos. [Quality of beans (*Phaseolus vulgaris* L.) stored in a silobag under field conditions in relation to broken kernels and foreign matter]. *Rev. la Fac. Agron. UNLP* 118, 1–8. <https://doi.org/10.24215/16699513e016>.
- Gregori, R., Meriggi, P., Pietri, A., Formenti, S., Baccarini, G., Battilani, P., 2013. Dynamics of fungi and related mycotoxins during cereal storage in silo bags. *Food Control* 30, 280–287. <https://doi.org/10.1016/j.foodcont.2012.06.033>.
- Hagstrum, D.W., 2001. Immigration of insects into bins storing newly harvested wheat on 12 Kansas farms. *J. Stored Prod. Res.* 37, 221–229. [https://doi.org/10.1016/S0022-474X\(00\)00023-0](https://doi.org/10.1016/S0022-474X(00)00023-0).
- Hagstrum, D.W., Phillips, T.W., Cuperus, G., 2012. *Stored Product Protection*. Kansas State University, Manhattan, Kansas, USA.
- Hidalgo, R., Pozzolo, O., Domínguez, F., Behr, E., Botta, G., 2012. Incidencia del diseño de embolsadoras en la calidad del arroz conservado en silo bolsa. [Incidence of bagging machine design on the quality of rice preserved in bag silo]. In: X Congreso Latinoamericano y Del Caribe de Ingeniería Agrícola e XLI Congresso Brasileiro de Engenharia Agrícola - CLIA/CONBEA 2012. Londrina - PR, Brasil, 15 a 19 de julho 2012, p. 7.
- Hidalgo, R., Domínguez, F., Botta, G., Pozzolo, O., Tourn, M., Ferrari, H., 2014. Detección temprana de la variabilidad en la calidad de semillas crudas de algodón (*Gossypium hirsutum*) conservadas en silo bolsas mediante mediciones de la concentración de CO₂. [Early detection of variability in the quality of cotton seeds (*Gossypium hirsutum*) stored in silo bags by CO₂ concentration measurement]. In: Cardoso, L., Bartosik, R., De La Torre, D., Abadía, M.B., Santa Juliana, M. (Eds.), *Almacenamiento de Granos en Silo Bolsas: Resultados de Investigación 2009-2013*. INTA, Balcarce, Argentina, pp. 48–56.
- Howlader, A.J., Matin, A.S.M.A., 1988. Observations on the pre-harvest infestation of paddy by stored grain pests in Bangladesh. *J. Stored Prod. Res.* 24, 229–231. [https://doi.org/10.1016/0022-474X\(88\)90024-0](https://doi.org/10.1016/0022-474X(88)90024-0).
- Idler, C., Wagner, A., Weber, U., Hoffmann, T., 2012. Effect of short-term storage on quality of wheat stored in large polyethylene bags. *Agric. Eng. Int. CIGR J.* 14, 149–156.
- Jian, F., Chelladurai, V., Jayas, D.S., Demianyk, C.J., White, N.D.G., 2014. Interstitial concentrations of carbon dioxide and oxygen in stored canola, soybean, and wheat seeds under various conditions. *J. Stored Prod. Res.* 57, 63–72. <https://doi.org/10.1016/j.jspr.2013.12.002>.
- Jian, F., Chelladurai, V., Jayas, D.S., White, N.D.G., 2015a. Three-dimensional transient heat, mass, and momentum transfer model to predict conditions of canola stored inside silo bags under Canadian prairie conditions: Part II. Model of canola bulk temperature and moisture content. *Trans. ASABE (Am. Soc. Agric. Biol. Eng.)* 58, 1135–1144. <https://doi.org/10.13031/trans.58.11053>.
- Jian, F., Chelladurai, V., Jayas, D.S., White, N.D.G., 2015b. Three-dimensional transient heat, mass, and momentum transfer model to predict conditions of canola stored inside silo bags under Canadian prairie conditions: Part I. Soil temperature model. *Trans. ASABE (Am. Soc. Agric. Biol. Eng.)* 58, 1127–1134. <https://doi.org/10.13031/trans.58.11052>.
- Magan, N., Lacey, J., 1984. Effects of gas composition and water activity on growth of field and storage fungi and their interactions. *Trans. Br. Mycol. Soc.* 82, 305–314. [https://doi.org/10.1016/s0007-1536\(84\)80074-1](https://doi.org/10.1016/s0007-1536(84)80074-1).
- Marcos Valle, F., Castellari, C., Yommi, A., Pereyra, M.A., Bartosik, R., 2021a. Evolution of grain microbiota during hermetic storage of corn (*Zea mays* L.). *J. Stored Prod. Res.* 92, 101788. <https://doi.org/10.1016/j.jspr.2021.101788>.
- Marcos Valle, F., Gastón, A., Abalone, R., de la Torre, D., Castellari, C., Bartosik, R., 2021b. Study and modelling the respiration of corn seeds (*Zea mays* L.) during hermetic storage. *Biosyst. Eng.* 8, 45–57. <https://doi.org/10.1016/j.biosystemseng.2021.05.009>.
- Marín, S., Companys, E., Sanchis, V., Ramos, A.J., Magan, N., 1998. Effect of water activity and temperature on competing abilities of common maize fungi. *Mycol. Res.* 102, 959–964. <https://doi.org/10.1017/S0953756297005613>.
- Marrugo-Ligardo, Y., Fuentes-Berrio, L., Montero-Castillo, P., Acevedo-Correa, D., 2015. Factibilidad de almacenamiento de semillas de ajonjolí (*sesamum indicum* L.) en bolsas silobag. [Feasibility of storage of sesame seeds (*Sesamum indicum* L.) in silobag bags]. *Biotechnol. en el Sect. Agropecu.* 13, 83–89.
- Massigoge, J., Cardoso, L., Bartosik, R., Rodríguez, J., Ochandío, D., 2010. Almacenamiento de cebada cervecera en silos bolsa. [Storage of maltin barley in silo-bags]. In: IX Congreso Latinoamericano y Del Caribe de Ingeniería Agrícola - CLIA 2010 y XXXIX Congresso Brasileiro de Engenharia Agrícola - CONBEA. Associação Brasileira de Engenharia Agrícola. Vitória, Brasil, July 25–29 2010.
- Matthews, S., Noli, E., Demir, I., Khajeh-Hosseini, M., Wagner, M.H., 2012. Evaluation of seed quality: from physiology to international standardization. *Seed Sci. Res.* 22, S69–S73. <https://doi.org/10.1017/S0960258511000365>.
- Moreno Ferro, S., Paturlanne, J., 2015. Costos y desempeño económico del traslado en el tiempo de productos agrícolas: el silobolsa en Argentina. [Costs and economic performance of the transfer over time of agricultural products: the silobolsa in Argentina]. *Rev. Investig. Agropecu.* 41, 325–330.
- Navarro, S., Donahaye, J.E., Fishman, S., 1994. The future of hermetic storage of dry grains in tropical and subtropical climates. In: Highley, E., Wright, E.J., Banks, H.J., Champ, B.R. (Eds.), *Proc. 6th International Working Conference on Stored-Product Protection*. CAB International, Wallingford, United Kingdom, pp. 130–138, 17–23 April 1994.
- Navarro, S., Timlick, B., Demianyk, C.J., White, N.D.G., 2012. Controlled or modified atmospheres. In: Hagstrum, D., Phillips, T., Cuperus, G. (Eds.), *Stored Product Protection*. Kansas State University, Manhattan, Kansas, USA, pp. 191–202.
- Navarro, S., Zettler, J., 2000. Critical limits of degree for sealing for successful application of controlled atmosphere or fumigation. In: Donahaye, E.J., Navarro, S., Leesch, J.G. (Eds.), *Proceedings of the 6th International Conference Controlled Atmosphere and Fumigation in Stored Products*. Executive Printing Services, pp. 507–520. Clovis, California, USA, Fresno, CA, USA.
- Ochandío, D., Bartosik, R., Gastón, A., Abalone, R., Arias Barreto, A., Yommi, A., 2017. Modelling respiration rate of soybean seeds (*Glycine max* (L.)) in hermetic storage. *J. Stored Prod. Res.* 74, 36–45. <https://doi.org/10.1016/j.jspr.2017.09.001>.
- Ochandío, D., Cardoso, L., Bartosik, R., de la Torre, D., Rodríguez, J., Massigoge, J., 2010. Storage of canola in hermetic plastic bags. In: Carvalho, M.O., Fields, P.G., Adler, C.S., Arthur, F.H., Athanassiou, C.G., Campbell, J.F., Fleurat-Lessard, F., Flinn, P.W., Hodges, R.J., Isikber, A.A., Navarro, S., Noyes, R.T., Riudavets, J., Sinha, K.K., Thorpe, G.R., Timlick, B.H., Trematerra, P., W, N.D.G. (Eds.), 10th International Working Conference on Stored Products Protection. Julius-Kun - Archiv, Estoril, Portugal, pp. 323–330. <https://doi.org/10.5073/jka.2010.425.167.321>. June 27 - July 2 of 2010.
- Ochandío, D., Massigoge, J., Castellari, C., 2014a. Almacenamiento hermético in vitro de granos de avena, canola, girasol, maíz, soja y sorgo. [In-vitro hermetic storage of oat, canola, sunflower, corn soybean and sorghum]. In: Gastón, A., Abalone, R., Bartosik, R. (Eds.), *Memorias Del Primer Congreso Internacional de Almacenamiento de Granos En Silo Bolsa*. INTA Ediciones, Mar del Plata/Balcarce, Argentina, ISBN 978-987-3662-06-5, pp. 159–167. October 13 - 16.
- Ochandío, D., Massigoge, J., Bartosik, R., Cardoso, L., 2014b. Almacenamiento de sorgo granífero húmedo en bolsas plásticas herméticas en el sur de la provincia de Buenos Aires. [Storage of wet sorghum in hermetic plastic bags (silo-bags) in south of Buenos Aires province]. In: En Gastón, A., Abalone, R., Bartosik, R. (Eds.), *Memorias Del Primer Congreso Internacional de Almacenamiento de Granos En Silo Bolsa*. INTA Ediciones, Mar del Plata/Balcarce, Argentina, pp. 133–139 (ISBN 978-987-3662-06-5), October 13 - 16.
- Pacin, A.M., Bovier, E.C., González, H.L., Whitechurch, E.M., Martínez, E.J., Resnik, S.L., 2009. Fungal and fumonisins contamination in Argentine maize (*Zea mays* L.) silo bags. *J. Agric. Food Chem.* 57, 2778–2781. <https://doi.org/10.1021/jf803609c>.
- Paster, N., Bullerman, L.B., 1988. Mould spoilage and mycotoxin formation in grains as controlled by physical means. *Int. J. Food Microbiol.* 7, 257–265. [https://doi.org/10.1016/0168-1605\(88\)90044-X](https://doi.org/10.1016/0168-1605(88)90044-X).
- Recendiz Hurtado, F., Góngora, F.J., Mena Mungía, S., Sánchez Martínez, J., 2011. Conservación de grano de maíz en el sistema de almacenamiento Silo-Bolsa en Guadalajara, Jalisco, México. [Storage of corn in the silo-bag system in Guadalajara, Jalisco, Mexico]. *scientia-CUCBA* 13, 44–50.
- Ridley, A.W., Burrill, P.R., Cook, C.C., Daglish, G., 2011. Phosphine fumigation of silo bags. *J. Stored Prod. Res.* 47, 349–356. <https://doi.org/10.1016/j.jspr.2011.06.001>.
- Rodríguez, J., Bartosik, R., Cardoso, L., Croce, D., 2008. Factors affecting carbon dioxide concentration in interstitial air of wheat stored in hermetic plastic bags (silo-bag). In: Daolin, G., Navarro, S., Jian, Y., Cheng, T., Zuxun, J., Yue, L., Yang, L., Haipeng, W. (Eds.), *Proceedings of the 8th International Conference Controlled Atmospheres and Fumigation of Stored Products*. Sichuan Publishing House of Science and Technology, Chengdu, China, pp. 589–592.
- Ruffato, S., Taffarel, C., Solange, Bonaldo, M., 2014. Quality evaluation of corn stored in silo bag. In: Gastón, A., Abalone, R., Bartosik, R. (Eds.), *Memorias Del Primer Congreso Internacional de Almacenamiento de Granos En Silo Bolsa*. INTA, Mar del Plata, Argentina, pp. 169–173 (ISBN 978-987-3662-06-5), July 2 to 4 of 2014.
- Santa Juliana, M., Cardoso, L., 2014. Componentes del sistema de silo bolsa. [Silo-bag system components]. In: Cardoso, L., Bartosik, R., de la Torre, D., Abadía, B., Santa Juliana, M. (Eds.), *Almacenamiento de Granos En Silo Bolsa: Resultados de Investigación 2009-2013*. Instituto Nacional de Tecnología Agropecuaria (INTA), Buenos Aires, Argentina, pp. 15–22.
- Sone, J., 1999. Carbon dioxide production in stored maize as affected by moisture content, level of broken corn and foreign materials and infestation by *Sitophilus zeamais* Motschulsky. *J. Asia Pac. Entomol.* 2, 133–141. [https://doi.org/10.1016/S1226-8615\(08\)60041-2](https://doi.org/10.1016/S1226-8615(08)60041-2).
- Steele, J.L., Saul, R.A., Hukill, W.V., 1969. Deterioration of shelled corn as measured by carbon dioxide production. *Trans. ASAE (Am. Soc. Agric. Eng.)* 12, 685–689.
- Subramanyam, B., Lakshmikantha, C., Campabadal, C., Lawrence, J., Cardoso, L., Maier, D.E., 2012. Evaluation of silo bags for temporary storage of wheat. In: Navarro, S., Banks, H.J., Jayas, D.S., Bell, C.H., Noyes, R.T., Ferizli, A.G., Emekci, M., Isikber, A.A., Alagundaram, K. (Eds.), *Proceedings of the 9th International Conference Controlled Atmospheres and Fumigation of Stored Products*. Antalya, Turkey, pp. 535–541.
- Taher, H., Urcola, H., Cardoso, L., Bartosik, R., 2014. Percepción de los productores sobre beneficios y problemáticas en el almacenamiento en silo bolsa. [Farmers perception about pros and cons in the silo-bag storage system]. In: Gastón, A., Abalone, R., Bartosik, R. (Eds.), *Memorias Del Primer Congreso Internacional de Almacenamiento de Granos En Silo Bolsa*. INTA, Mar del Plata, Argentina, ISBN 978-987-3662-06-5, pp. 35–42.
- Taffarel, C., Ruffato, S., Danao, M.-G.C., Nogueira, R.M., Zandonadi, R.S., 2013. Storage of corn in silo bag: internal temperature profile and product quality, in: 2013 Kansas City, Missouri, July 21. American Society of Agricultural and Biological Engineers, St. Joseph, MI, pp. 2348–2357. <https://doi.org/10.13031/aim.20131596115>. (Accessed 24 July 2013).

- Taher, H., Urcola, H.A., Bartosik, R., Cendoya, M.G., 2019. Caracterización del uso del silo bolsa en la provincia de Buenos Aires [Characterization of the silo bag use by farmers in Buenos Aires province]. *Rev. Fac. Agron Univ Nac Rosario* 33, 25–30.
- Taher, Hernán, Urcola, H.A., Cendoya, M.G., Bartosik, R., 2019. Predicting soybean losses using carbon dioxide monitoring during storage in silo bags. *J. Stored Prod. Res.* 82, 1–8. <https://doi.org/10.1016/j.jspr.2019.03.002>.
- Thompson, T.L., 1972. Temporary storage of high-moisture shelled corn using continuous aeration. *Trans. ASAE (Am. Soc. Agric. Eng.)* 15, 333–337.
- Trematerra, P., 2015. Adult dispersal of *Sitotroga cerealella* in a conventional small-farm in Southern Italy. *Bull. Insectol.* 68, 111–118.
- Urcola, H., Bartosik, R., 2020. Storing soybeans in silo bags: economic outcomes and grain loss risk. *Rev. la Fac. Ciencias Agrar. la Univ. Nac. Cuyo* 1–17, 1853–8665.
- Ward, J., Davis, J., 2013. A system to assess grain bag storage internal environment. *Trans. ASABE (Am. Soc. Agric. Biol. Eng.)* 56, 1503–1509.
- Weinberg, Z.G., Yan, Y., Chen, Y., Finkelman, S., Ashbell, G., Navarro, S., 2008. The effect of moisture level on high-moisture maize (*Zea mays* L.) under hermetic storage conditions-in vitro studies. *J. Stored Prod. Res.* 44, 136–144. <https://doi.org/10.1016/j.jspr.2007.08.006>.
- Zufiaurre, E., Abba, A., Bilenca, D., 2020. Damage to silo bags by mammals in agroecosystems: a contribution for mitigating human-wildlife conflicts. *Wildl. Res. Early view*. <https://doi.org/10.1071/WR20045>.
- Zufiaurre, E., Abba, A., Bilenca, D., 2018. Assessment of stakeholder perceptions of the damage to silo bags by vertebrate species in Argentina. *Hum. Dimens. Wildl.* 24, 1–7. <https://doi.org/10.1080/10871209.2019.1538434>.